Glossary

Every storage-engine, systems, and concurrency term in this doc, alphabetical. Defining any of them in one sentence is part of the final exam.

Term	Plain-English meaning	Where it shows up in StrataKV
Acknowledged write	A write whose `Put` call has returned OK to the caller; the promise that it will survive a crash	The core guarantee
Acquire / release	Memory-ordering rules that make one thread's writes visible to another in the right order	Skip list
Arena	A big block of memory handed out in pieces and freed all at once	Memtable nodes
Atomic (operation)	Happens completely or not at all; no other thread sees it half-done	`std::atomic`, rename
Atomic rename	Renaming a file replaces the target in one step; readers see the old file or the new one, never a mix	`CURRENT` file
B-tree	A sorted tree of fixed-size pages, updated in place; used by SQLite and most databases	The main alternative to LSM
Benchmark	A repeatable measurement of performance	P6
Bloom filter	A compact bit array that answers "definitely not here" or "maybe here"	One per SSTable
Bottom level	The last level in the LSM tree (L1 here); nothing older lies below it	Where tombstones can be dropped
Checksum (CRC32C)	A small fingerprint of data used to detect corruption or partial writes	WAL records, SSTable footer

Term	Plain-English meaning	Where it shows up in StrataKV
Compaction	Merging SSTables to remove overwritten and deleted data and keep reads fast	P3
Copy-on-write	Never modify data in place; write a new copy and switch a pointer to it	LMDB, BoltDB (alternative design)
Crash consistency	After any crash, the data on disk is a valid state that includes every acknowledged write	The fuzzer proves this
Crash point	A named place in the code where a test can force the process to die	`STRATAKV_CRASH_AT`
Data race	Two threads touching the same memory at once, at least one writing, with no synchronization; undefined behavior in C++	What TSan detects
Durability	Data survives a crash or power loss once acknowledged	`fsync` policy
False positive (bloom filter)	The filter says "maybe" for a key that isn't there; costs one wasted read	About 1% at 10 bits/key
`fsync`	A system call that forces a file's data from the OS cache onto the physical disk	Every durable write
`F_FULLFSYNC`	The macOS call that also flushes the drive's own cache, which plain `fsync` there does not	`Env::SyncFile` on macOS
Fault injection	Deliberately causing failures to test recovery	Crash fuzzer
Footer	Fixed-size block at the end of an SSTable pointing to its index and filter	SSTable format

Term	Plain-English meaning	Where it shows up in StrataKV
GIL	CPython's global interpreter lock: one thread runs Python bytecode at a time	pybind11 bindings
Group commit	Batching many writers' records into one write and one `fsync`	P4
Immutable memtable	A full memtable frozen and waiting to be flushed to disk	Write path
Index block	A sorted list of each data block's last key and position, for finding the right block fast	SSTable format
Internal key	User key plus sequence number plus type; the form keys take inside the engine	Snippet 4
K-way merge	Merging k sorted lists into one, usually with a min-heap	Compaction
L0 / L1	Levels of SSTables: L0 holds fresh flushes that may overlap; L1 holds compacted, non-overlapping files	Read path
Leader-follower	One thread does shared work on behalf of waiting threads, then hands off the leader role	Group commit
Leveled compaction	Organizing SSTables into levels with size limits and merging downward	P3
LSM tree	Log-structured merge tree: buffer writes in memory, flush sorted files, merge them in the background	The whole design
MANIFEST	Append-only log recording which SSTables are live at each level	Version tracking
Memtable	The in-memory sorted structure holding recent writes	Skip list

Term	Plain-English meaning	Where it shows up in StrataKV
Model-based testing	Running random operations against both the real system and a simple model, and checking they agree	Hypothesis stateful tests
Property-based testing	Generating many random inputs to check a property always holds, then shrinking failures to the smallest case	Hypothesis
RAII	Tying a resource's lifetime to an object's scope so cleanup is automatic	Locks, files
Read amplification	How many places a lookup must check	Bloom filters reduce its cost
Recovery	Rebuilding in-memory state from disk on startup	WAL replay
Reference counting	Freeing an object when its count of owners reaches zero	`shared_ptr<Version>`
Sanitizer	A compiler mode that adds runtime checks for memory errors, undefined behavior, or races	`asan`, `tsan` presets
Sequence number	A counter that increases with every write, so newer versions are recognizable	Internal key
Shrinking	Hypothesis reducing a failing test case to the smallest one that still fails	Debugging
SIGKILL	A signal that ends a process instantly with no cleanup	Crash fuzzer
Size-tiered compaction	Merging files of similar size; lower write amplification, more disk use	Alternative to leveled

Term	Plain-English meaning	Where it shows up in StrataKV
Skip list	A sorted linked list with extra "express lane" pointers, giving fast search and easy concurrent reads	Memtable
Snapshot	A consistent view of the data as of one moment	Version references
Space amplification	Disk used divided by live data size	Compaction reduces it
SSTable	Sorted String Table: an immutable file of sorted key-value pairs with an index	On-disk format
Tombstone	A marker recording that a key was deleted	Deletes
Torn write	A write cut off partway by a crash, leaving a partial record	WAL tail
Varint	Variable-length integer encoding; small numbers take fewer bytes	Lengths in records
Version (VersionSet)	The set of live SSTables at a moment; readers hold one so its files aren't deleted	Snippet 7
WAL (write-ahead log)	An append-only log where every write is recorded before it's applied, so it can be replayed after a crash	P1
Write amplification	Bytes physically written to disk divided by bytes the user wrote	Compaction cost
Write batch	A group of puts and deletes applied atomically	`WriteBatch`